

Exercise 1: Control Structures

Table used

Let's use the schema from your exercise. [PLSQL Exercises](#)

Customers Table

CustomerID	Name	DOB
1	John	1960-05-15
2	Jane	1990-07-20

Loans Table

LoanID	CustomerID	InterestRate
101	1	5
102	2	7

Scenario 1: The bank wants to apply a discount to loan interest rates for customers above 60 years old.

- **Question:** Write a PL/SQL block that loops through all customers, checks their age, and if they are above 60, apply a 1% discount to their current loan interest rates.

```
for c IN (select customer_id, dob from customers)
loop
    if ( trunc(months_between(sysdate, c.dob)/12)>60 )
    then
        update Loans l
        set InterestRate = InterestRate - 1
        where l.customer_id = c.customer_id;
    end if
end loop
end;
```

Scenario 2: A customer can be promoted to VIP status based on their balance.

- **Question:** Write a PL/SQL block that iterates through all customers and sets a flag IsVIP to TRUE for those with a balance over \$10,000.

Table used:

Now the table becomes:

CustomerID	Name	DOB	Balance	LastModified	IsVIP
1	John	...	15000	...	NULL
2	Jane	...	5000	...	NULL

```
for c in (select customer_id, balance, isVIP from customers c)
loop
  if c.balance > 10000 then
    update customers
    set isVIP = 'true'
    where customer_id = c.customer_id;
  end if
end loop
end;
```

Scenario 3: The bank wants to send reminders to customers whose loans are due within the next 30 days.

- **Question:** Write a PL/SQL block that fetches all loans due in the next 30 days and prints a reminder message for each customer.

Schema used:

Customers Table

CustomerID	Name
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1	John
---	------

2	Jane
---	------

Loans Table

LoanID	CustomerID	EndDate
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101	1	10-Jul-2026
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102	2	25-Dec-2026
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```
BEGIN
for c in (select c.customerID ,c.Name, l.EndDate from customers c
          inner join Loans l
          on l.customerID = c.customerID)
loop
IF c.EndDate BETWEEN SYSDATE AND SYSDATE + 30 THEN      dbms_output.put_line(
    'reminder message your loan should repiad within'
    || round(c.EndDate - sysdate));
    end if;
end loop;
end;
```